Arian Merati

Arian Merati | arian.merati@gmail.com

Education

Informatics MInf: Computer Science, **First Class** Honours, Expected in 06/2027
University of Edinburgh - Edinburgh, EDH

Research Paper Contributions

REFRAMED: Towards Realistic Audio Description Generation for Movies COLM 2026
In the acknowledgements section for the ASR evaluation pipeline and multimodal dataset alignment

Experience

Junior Research Assistant, University of Edinburgh, 09/2026 - Present

- Conducting research on reasoning in LLM's and on-policy distillation, jointly supervised by Professor Ivan Titov and Professor Shay Cohen
- Investigating what teachers distributions and data, causes on-policy distillation methods to learn efficiently.
- Developing a theoretical framework for what causes on-policy distillation to work well in LLMs

Junior Research Assistant, University of Edinburgh, 06/2025 - Present

- Conducting research on improving LLM's on long form writing tasks, focusing on story writing, supervised by Professor Mirella Lapata
- Researching on-policy self-distillation methods for creative writing; lead author of a manuscript in preparation for ACL 2027
- Investigating multi-agent systems and reinforcement learning methods to improve reasoning capabilities in creative writing tasks, in order to generate more creative stories.

Research Intern, University of Edinburgh, 01/2025 - 06/2025

- Conducted research at the Edinburgh Language and Vision Group supervised by Professor Frank Keller, focusing on automatic generation of audio descriptions
- Contributed to the evaluation pipeline and aligned training dataset to *REFRAMED: Towards Realistic Audio Description Generation for Movies*, published at COLM 2026
- Created a specialized evaluation pipeline for automatic audio description transcriptions, including metrics such as Timestamp Accuracy, Character Name Error Rate, DER, WER (evaluated on both diarization and gold-standard timestamps), Sentence Error Rate
- Benchmarked the performance of various automatic speech recognition systems, such as WhisperX and Speechmatics, using the specialized evaluation pipeline
- Aligned audio descriptions from the AudioVault dataset with video clips from the Condensed Movies dataset to create a synchronized multimodal training dataset.

Research Intern, University of Edinburgh, 01/2024 - 11/2024

- Conducted research at the MAPS lab supervised by Professor Chris Xiaoxuan Lu
- Worked on thermal camera based 3D Occupancy Prediction, which aims to improve the performance of occupancy prediction in adverse weather conditions
- Generated baseline results by adapting RGB occupancy prediction models, such as GaussianFormer, to work with the NTU4DRadLM thermal dataset.

Research Intern, University of Surrey, 06/2023 - 09/2023

- Conducted research at the CAV lab, supervised by Professor Saber Fallah, working on deep learning and autonomous vehicles.
- Replaced the VGG-16 backbone of NetVLAD with a Vision Transformer backbone and pre-trained it, using the BYOL framework.
- Modified the DeCLIP framework at a structural level so that it can accept two image inputs rather than the standard image–text pairing,
- Developed a YOLOv5-based pipeline to generate heat-map images and aligned raw sensor data, contributing to the construction of datasets for autonomous vehicle perception research

Other Qualifications

A-Levels - A* Maths, A* Further Maths, A* Physics, A Chemistry